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UPES
UNIVERSITY OF TOMORROW

Minor Project-II

Title: SmartEvaluate

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1. Abstract

The **Automated Answer Sheet Evaluation System** using **Handwritten Text Recognition (HTR)** aims to automate the process of evaluating handwritten answer sheets. The traditional method of manual evaluation is time-consuming and error-prone. This system leverages the power of machine learning and optical character recognition (OCR) techniques, specifically HTR, to assess and grade answer sheets efficiently and accurately. The project will involve building a model that can recognize and interpret handwritten answers, and evaluate them based on predefined criteria.

2. Introduction

In educational institutions, the process of evaluating answer sheets manually is a labor-intensive task, requiring a lot of human effort and time. With the rise of modern technologies, the need for automation in this field has become evident. Handwritten Text Recognition (HTR) has emerged as a solution for automating the reading and interpretation of handwritten documents, especially in answering examination sheets. This project focuses on utilizing HTR to develop an **Automated Answer Sheet Evaluation System** that can recognize, evaluate, and score handwritten answers, reducing the burden on educators while maintaining accuracy and efficiency.

3. Literature Review

- Handwritten Text Recognition (HTR): HTR is a subfield of Optical Character Recognition (OCR) that deals with converting handwritten text into machine-readable formats. Recent advancements in deep learning models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have significantly improved the accuracy of HTR systems.
- Automated Evaluation Systems: Previous research has shown that automated grading systems, particularly those for multiple-choice or fill-in-the-blank questions, have been successfully implemented. However, the challenge remains in accurately evaluating open-ended, subjective answers in essay-type questions.

4. Problem Statement

Manual evaluation of handwritten answer sheets is a time-consuming and resource-intensive process. Moreover, human evaluators are prone to errors and biases. The existing automated grading systems are generally limited to objective-type questions and do not handle subjective questions well. This project aims to develop a system that can automatically evaluate handwritten subjective answers with high accuracy using HTR technology.

5. Objectives

- To design an Automated Answer Sheet Evaluation System that can read and assess handwritten answers accurately.

- To reduce the time spent by educators on grading.

6. Methodology

- Data Collection: The system will be trained using datasets of handwritten text from various subjects and different handwriting styles.

- Preprocessing: Handwritten answers will be preprocessed to remove noise and enhance text recognition.

- Handwritten Text Recognition (HTR): Deep learning models, such as CNN and LSTM (Long Short-Term Memory networks), will be used for HTR to convert handwritten answers into digital text.

- Evaluation and Grading: The system will evaluate answers based on predefined criteria, such as relevance, completeness, and accuracy, using NLP techniques.

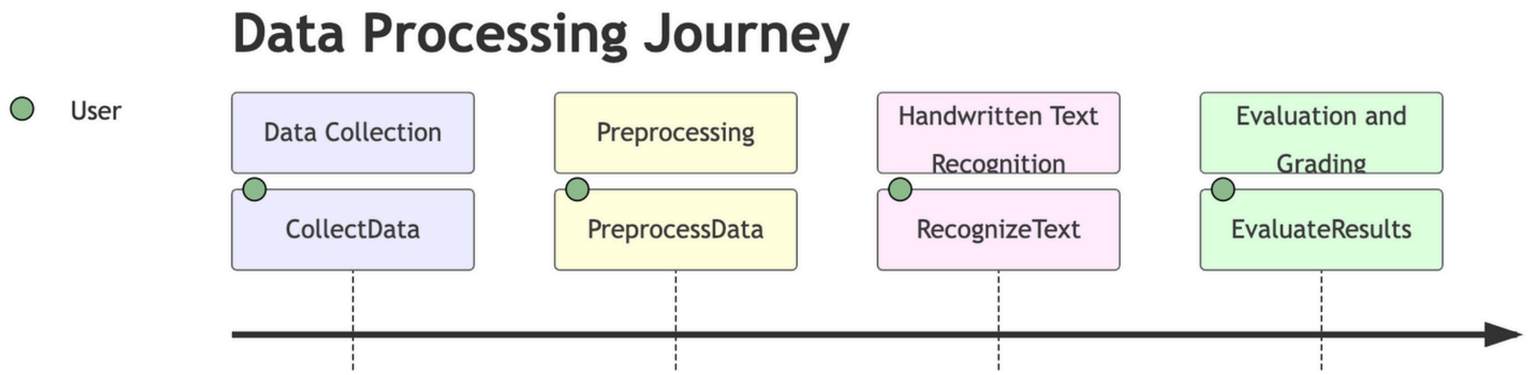


Diagram Source:Self

7.

Gantt Chart

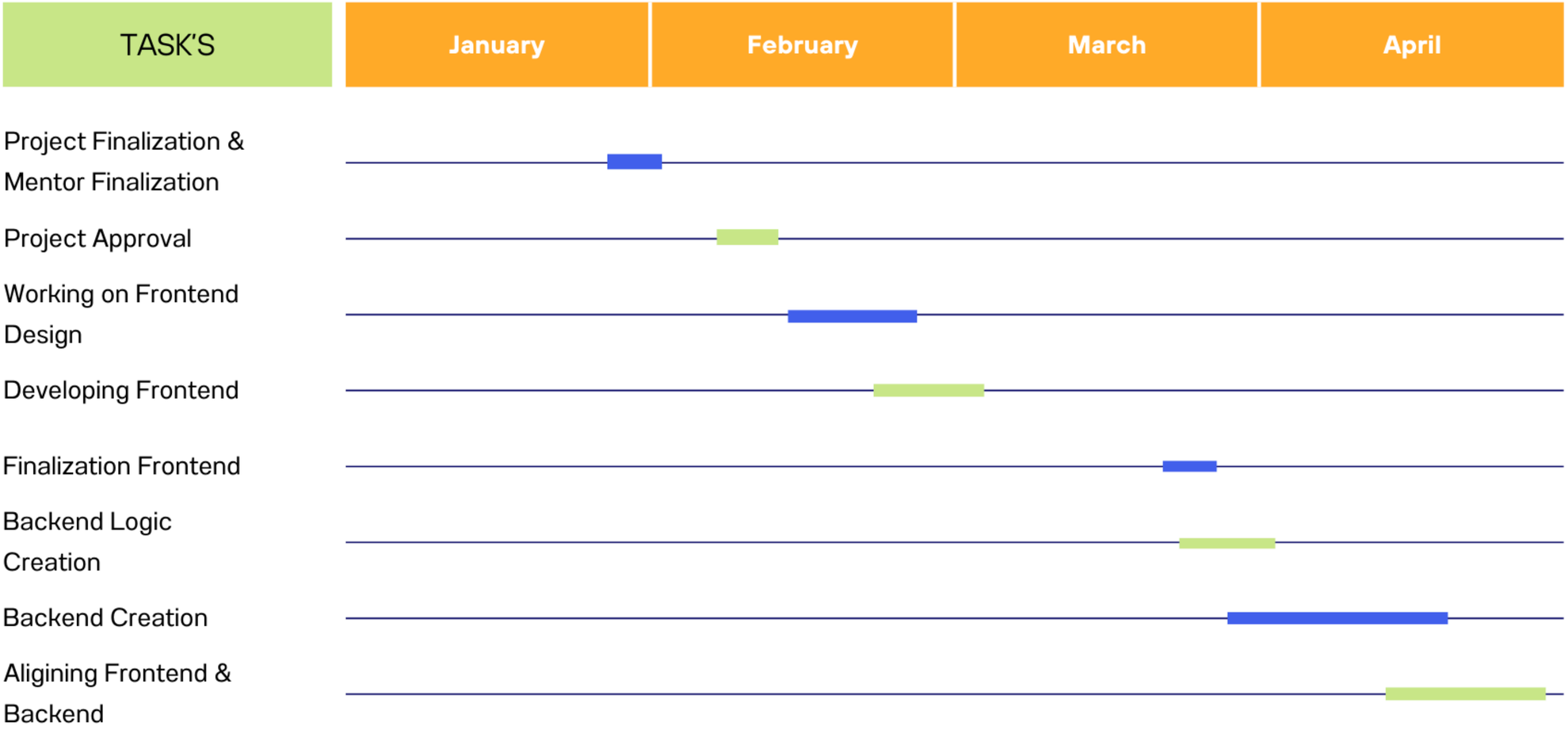


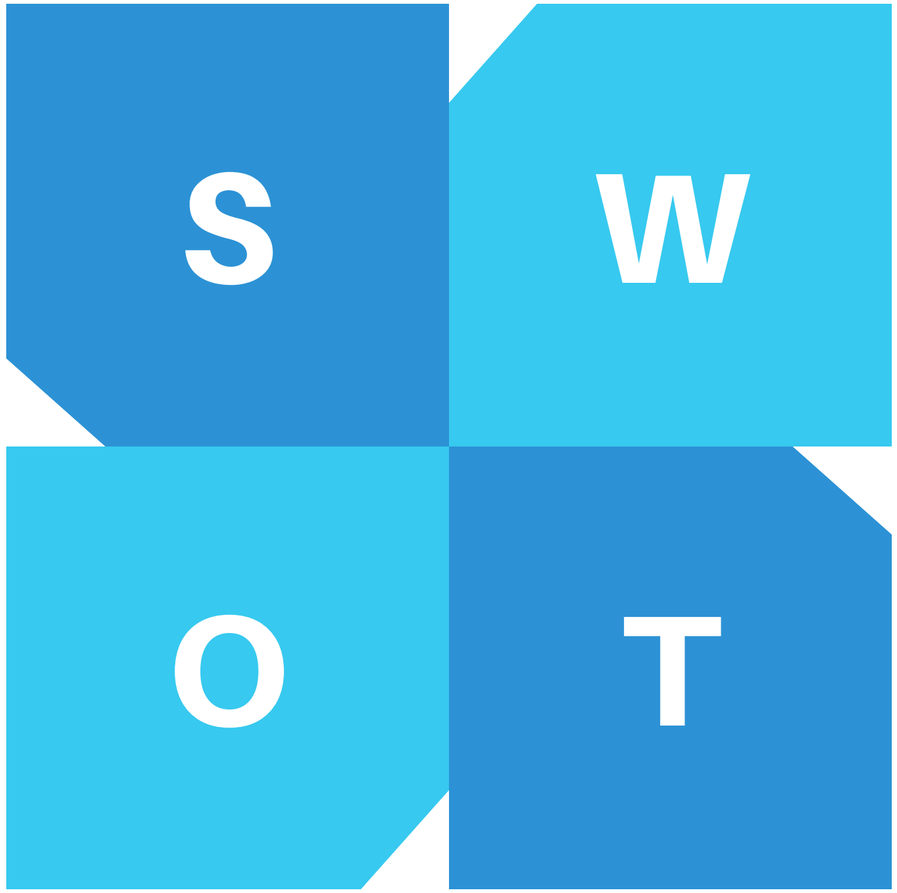
Diagram Source:Self

STRENGTHS

- Accuracy & Consistency – Reduces human errors and ensures objective grading.
- Efficiency & Speed – Provides instant feedback, saving time for educators.

OPPORTUNITIES

- Integration with Learning Management Systems (LMS) – Can be embedded into digital education platforms



WEAKNESSES

- Initial Implementation Cost – High setup costs for software, hardware, and training.
- Dependenceon Handwriting Recognition (HTR) Accuracy – May struggle with poor handwriting or diverse writing styles.

THREATS

- Resistance to Change
- Data Security & Privacy Concerns

8. Conclusion

By means of Handwritten Text Recognition (HTR) and Machine Learning, SMART EVALUATE transforms academic tests, therefore solving manual evaluation issues including time consumption, mistakes, and scalability. Using sophisticated image processing, deep learning, semantic matching, and dynamic scoring, it automates evaluation of handwritten answer sheets.

9. References

- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- OpenAI (2021). “Transformers for Optical Character Recognition.”
- Smith, R. (2007). “An Overview of the Tesseract OCR Engine,” ICDAR.
- ISO/IEC 19757-3:2006. Handwriting Recognition Standard.
- Online resources on OCR and HTR technologies.
- Articles and technical blogs on Handwritten Text Recognition (HTR), Optical Character Recognition (OCR), and AI in education from sources like Medium, Towards Data Science, and official documentation (TensorFlow, PyTorch, OpenCV).

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Thank You